

backend/package.json

```
{
  "name": "kaaba-backend",
  "version": "1.0.0",
  "type": "module",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "cors": "^2.8.5",
    "dotenv": "^16.4.5",
    "express": "^4.19.2",
    "ffmpeg-static": "^5.2.0",
    "fluent-ffmpeg": "^2.1.3",
    "openai": "^4.57.0",
    "socket.io": "^4.7.5",
    "socket.io-client": "^4.7.5",
    "yt-dlp-exec": "^1.0.2"
  }
}
```

backend/server.js

```
import express from "express";
import cors from "cors";
import dotenv from "dotenv";
import OpenAI from "openai";
import { createServer } from "http";
import { Server } from "socket.io";
import ffmpeg from "fluent-ffmpeg";
import ffmpegPath from "ffmpeg-static";
import fs from "fs";
import ytDlp from "yt-dlp-exec";

ffmpeg.setFfmpegPath(ffmpegPath);

dotenv.config();

const app = express();
```

```

const server = createServer(app);

const io = new Server(server, {
  cors: {
    origin: "*",
  },
});

app.use(cors());
app.use(express.json());

const openai = new OpenAI({
  apiKey: process.env.OPENAI_API_KEY,
});

app.get("/", (req, res) => {
  res.send("Kaaba Translator Backend Running");
});

app.post("/api/start", async (req, res) => {
  const { youtubeUrl } = req.body;

  if (!youtubeUrl) {
    return res.status(400).json({ error: "Missing YouTube URL" });
  }

  res.json({ success: true });

  try {
    const audioFile = "audio.mp3";

    await ytDlp(youtubeUrl, {
      extractAudio: true,
      audioFormat: "mp3",
      output: audioFile,
      noPlaylist: true,
    });

    const segmentFolder = "segments";

    if (!fs.existsSync(segmentFolder)) {
      fs.mkdirSync(segmentFolder);
    }

    ffmpeg(audioFile)
      .outputOptions([
        "-f segment",
        "-segment_time 20",

```

```

        "-c copy",
    ])
    .output(`${segmentFolder}/output%03d.mp3`)
    .on("end", async () => {
        const files = fs.readdirSync(segmentFolder);

        for (const file of files) {
            const transcription = await openai.audio.transcriptions.create({
                file: fs.createReadStream(`${segmentFolder}/${file}`),
                model: "whisper-1",
                language: "ar",
            });

            const translation = await openai.chat.completions.create({
                model: "gpt-4o-mini",
                messages: [
                    {
                        role: "system",
                        content:
                            "Translate Arabic Islamic speech into natural English.",
                    },
                    {
                        role: "user",
                        content: transcription.text,
                    },
                ],
            });

            const translatedText =
                translation.choices[0].message.content;

            io.emit("translation", {
                arabic: transcription.text,
                english: translatedText,
            });
        }
    })
    .run();
} catch (error) {
    console.error(error);
}
});

io.on("connection", (socket) => {
    console.log("Client connected");
});

const PORT = process.env.PORT || 3001;

```

```
server.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

backend/.env

```
OPENAI_API_KEY=your_openai_api_key  
PORT=3001
```

frontend/package.json

```
{  
  "name": "frontend",  
  "private": true,  
  "version": "1.0.0",  
  "scripts": {  
    "dev": "vite",  
    "build": "vite build",  
    "preview": "vite preview"  
  },  
  "dependencies": {  
    "axios": "^1.7.2",  
    "react": "^18.3.1",  
    "react-dom": "^18.3.1",  
    "socket.io-client": "^4.7.5"  
  },  
  "devDependencies": {  
    "@vitejs/plugin-react": "^4.3.1",  
    "vite": "^5.4.21"  
  }  
}
```

frontend/src/main.jsx

```
import React from "react";  
import ReactDOM from "react-dom/client";
```

```
import App from "./App";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

frontend/src/App.jsx

```
import { useEffect, useState } from "react";
import axios from "axios";
import { io } from "socket.io-client";

const socket = io("https://YOUR-RAILWAY-URL.up.railway.app");

export default function App() {
  const [youtubeUrl, setYoutubeUrl] = useState(
    "https://www.youtube.com/watch?v=4i4A7cZkKu4"
  );

  const [arabic, setArabic] = useState("");
  const [english, setEnglish] = useState("");
  const [loading, setLoading] = useState(false);

  useEffect(() => {
    socket.on("translation", (data) => {
      setArabic(data.arabic);
      setEnglish(data.english);
    });

    return () => {
      socket.off("translation");
    };
  }, []);

  const startTranslation = async () => {
    setLoading(true);

    try {
      await axios.post("/api/start", {
        youtubeUrl,
      });
    } catch (error) {
```

```

        console.error(error);
    }
};

return (
    <div
        style={{
            minHeight: "100vh",
            background: "black",
            color: "white",
            padding: "40px",
            fontFamily: "Arial",
        }}
    >
        <h1>Kaaba Live Translator</h1>

        <input
            value={youtubeUrl}
            onChange={(e) => setYoutubeUrl(e.target.value)}
            placeholder="Paste YouTube livestream"
            style={{
                width: "100%",
                padding: "12px",
                marginTop: "20px",
            }}
        />

        <button
            onClick={startTranslation}
            style={{
                marginTop: "20px",
                padding: "12px 24px",
                cursor: "pointer",
            }}
        >
            {loading ? "Starting..." : "Start Translation"}
        </button>

        <div
            style={{
                marginTop: "40px",
                background: "#111",
                padding: "20px",
                borderRadius: "10px",
            }}
        >
            <h2>Arabic</h2>
            <p>{arabic}</p>

```

```
        <h2>English Translation</h2>
        <p>{english}</p>
    </div>
</div>
);
}
```

frontend/vite.config.js

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react";

export default defineConfig({
  plugins: [react()],
  server: {
    port: 5173,
  },
});
```

vercel.json

```
{
  "rewrites": [
    {
      "source": "/api/(.*)",
      "destination": "https://YOUR-RAILWAY-URL.up.railway.app/api/$1"
    }
  ]
}
```

IMPORTANT

Replace:

YOUR-RAILWAY-URL

with your actual Railway backend URL in:

- App.jsx
- vercel.json

DEPLOYMENT

Backend (Railway)

Deploy:

backend/

Add environment variable:

OPENAI_API_KEY

Frontend (Vercel)

Root directory:

frontend

Framework preset:

Vite

INSTALL COMMANDS

Backend

```
npm install
```

Frontend

```
npm install
```

GIT PUSH

```
git add .  
git commit -m "kaaba live translator"  
git push
```